Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Projective measurements are equilibration processes

Files.

- File: `briefs/ProjectiveMeasurement/ProjectiveMeasurement.lean`
- Library: `ProjectiveMeasurement`
- Imports:
  `ForPhyslib.QuantumMechanics.Dephasing`,
  `ForPhyslib.QuantumMechanics.RelativeEntropy`,
  and the brief `UnitaryEntropy` (for `vonNeumann`)
- New files in `ForPhyslib/QuantumMechanics/`: `Pinching` (spectral projectors as the functional calculus of indicators, the pinching, the commutant, trace duality, density operators and supports, the Hilbert–Schmidt Pythagoras identity), `Dephasing` (the Lindblad generator, the explicit solution, uniqueness, the limit), `RelativeEntropy` (relative entropy, Klein's inequality with its equality case, the relative-entropy Pythagoras identity). They use from TauCeti:
  - the Hilbert–Schmidt norm, `TauCeti.Analysis.InnerProductSpace.HilbertSchmidt.Basic`;

  - finiteness of the spectrum, `TauCeti.Analysis.Normed.Operator.FiniteDimensional`;
  - the eigenvector action of the functional calculus, `TauCeti.Analysis.InnerProductSpace.FunctionalCalculusComparison`;
  - the supporting-line bound for $x \log x$, `TauCeti.Analysis.SpecialFunctions.Log.MulLog`.

- New `ForMathlib` material: `ForMathlib/Analysis/Calculus/MeanValue.lean` (a function continuous from the right at a with derivative zero on (a, ∞) is constant on $[a, \infty)$).

Setting. E is a finite-dimensional complex Hilbert space with density operators as in the brief "Schrödinger equation is equivalent to information conservation". A process is a map Φ sending an observable X and an incoming ensemble ρ to $\Phi(X, \rho)$; the conditions constrain it for self-adjoint X and density operators ρ. For self-adjoint X, `spectralProj` X x is `cfc` of the indicator of $\{x\}$, the projector P_x onto the x-eigenspace, and `pinching` X $\rho = \sum_x P_x \rho P_x$, the sum running over the real spectrum of X. `dephasingGenerator` X $\rho = X\rho X - \frac{1}{2}(X^2\rho + \rho X^2)$. The Hilbert–Schmidt norm is TauCeti's `hilbertSchmidtNorm`. The relative entropy `relEntropy` ρ $\sigma \in [0, \infty]$ is $\operatorname{tr}(\rho \log \rho) - \operatorname{tr}(\rho \log \sigma)$ if $\ker \sigma \subseteq \ker \rho$ and ∞ otherwise, with $\log =$ `cfc Real.log`, which is the logarithm on the support since $\log 0 = 0$. `IsUniqueMinOn` f s a says that $a \in s$ and $f(a) < f(b)$ for every other $b \in s$; `commutingDensities` X is the set of density operators commuting with X. The conditions are

- `EqMeas`: $\Phi(X, \rho) = P_X(\rho)$;
- `EqLind`: every $r : \mathbb{R} \to \operatorname{End}(E)$ with $r(0) = \rho$, $r'(t) = L(r(t))$ for $t > 0$ and r continuous from the right at 0 tends to $\Phi(X, \rho)$;
- `EqHS`: $\Phi(X, \rho)$ is the unique minimizer of $\|\rho - \sigma\|_{HS}$ over `commutingDensities` X;
- `EqKL`: $\Phi(X, \rho)$ is the unique minimizer of $D(\rho \| \sigma)$ over the same set.

Results.

Brief	Lean
Eqs. (3)–(4), blockwise evolution	`spectralProj_mul_dephasingGenerator_mul`, `dephasingGenerator_block`
Eq. (5), the solution	`dephasing`, `hasDerivAt_dephasing`, `dephasing_zero`; uniqueness `eq_dephasing`
Eqs. (6)–(10), the limit	`tendsto_dephasing`, `tendsto_pinching_of_hasDerivAt`
EQ-LIND ⇔ EQ-MEAS	`eqMeas_iff_eqLind`
"$[\sigma, X] = 0$ iff $P_X(\sigma) = \sigma$"	`commute_iff_pinching_eq`
P_X is positive and trace preserving	`isDensity_pinching`
Eq. (11)	`sq_hilbertSchmidtNorm_sub`, `hilbertSchmidtNorm_sub_pinching_lt`
EQ-HS ⇔ EQ-MEAS	`eqMeas_iff_eqHS`
"$\operatorname{tr}(\rho \log \sigma) = \operatorname{tr}(P_X(\rho) \log \sigma)$"	`trace_pinching_mul_of_commute`
Eq. (12)	`relEntropy_eq_add`
Eq. (13)	`relEntropy_pinching`, `relEntropy_pinching_eq`
"non-negative and vanishes iff $\sigma = P_X(\rho)$" (Klein)	`re_trace_mul_log_sub_log_nonneg`, `re_trace_mul_log_sub_log_eq_zero_iff`, `relEntropy_eq_zero_iff`
Supports in eq. (12)	`ker_pinching_le`, `ker_le_ker_pinching`
EQ-KL ⇔ EQ-MEAS	`eqMeas_iff_eqKL`, via `relEntropy_pinching_lt`
Theorem 1, complete	`projective_measurement_tfae`

Differences from the brief.

- Finite dimension only; the brief has a separable Hilbert space. Then every self-adjoint X has pure point spectrum, every density operator has finite entropy, so [B]FINENT holds and the two halves of Theorem 1 merge into one equivalence of four conditions. The convergence in EQ-LIND is in operator norm, which in finite dimension is trace-norm convergence; the non-commutative Scheffé lemma and dominated convergence are not needed.

- The Lindblad equation is assumed only for $t > 0$, with continuity from the right at $t = 0$. Under these assumptions the solution is unique (`eq_dephasing`), which the brief takes for granted when it solves eq. (4). EQ-LIND is stated for every solution; the explicit one shows that there is one.

- The Hilbert–Schmidt Pythagoras identity and the strict inequality hold for every operator σ commuting with X, not only for density operators; the constraint $\sigma \in D(H)$ in EQ-HS is not needed. For EQ-KL it is, through Klein's inequality.

- Eq. (13) is stated with the entropy `vonNeumann` of the brief "Schrödinger equation is equivalent to information conservation" and the identity $S(\rho) = -\operatorname{tr}(\rho \log \rho)$ (`vonNeumann_eq_neg_re_trace`). As a value in $[0, \infty]$ it is $\max(S(P_X(\rho)) - S(\rho), 0)$, which equals $S(P_X(\rho)) - S(\rho)$ by Klein's inequality.

- The relative entropy takes values in $[0, \infty]$, as physlib's `qRelativeEnt`. physlib's pinching and `pinching_pythagoras` are stated for matrices and pinch by the eigenprojectors of a state; this formalization works with operators on E, as the other briefs do, and pinches by an observable.

Problems found in the brief. No errors in the statements. Gaps and implicit points:

- Klein's inequality, "the last term is non-negative and vanishes if and only if $\sigma = P_X(\rho)$", is used without a reference. It is proved here in eigenbases: with $c_{ij} = |\langle u_i|v_j\rangle|^2$, $D(\rho\|\sigma) = \sum_{ij} c_{ij}(r_i \log r_i - r_i \log s_j - r_i + s_j) \geq \sum_{ij} c_{ij}(\sqrt{r_i} - \sqrt{s_j})^2$.
- "For non-faithful σ, the identity is understood on its support" hides two support facts that eq. (12) needs: $\ker P_X(\rho) \subseteq \ker \rho$ for positive ρ, and $\ker \sigma \subseteq \ker \rho \Rightarrow \ker \sigma \subseteq \ker P_X(\rho)$ for σ commuting with X. The first also gives eq. (12) when $\ker \sigma \not\subseteq \ker \rho$: then both sides are infinite.
- "Both $\rho(t)$ and $P_X(\rho(0))$ are density operators": that $\rho(t)$ stays a density operator is asserted, not shown. It holds because the matrix $\left(e^{-t(x-y)^2/2}\right)_{x,y}$ is positive semidefinite, so by the Schur product theorem the blockwise multiplication preserves positivity. In infinite dimension the Scheffé step needs it; in finite dimension it is not needed and is not formalized.
- The sentence "This base condition states the ensembles have finite entropy" follows QSTO:H, which does not say this; it describes [B]FINENT, stated next.
- The remark on observables with continuous spectrum is not formalized; in finite dimension there is none.

1.1 Hilbert spaces of any dimension

This subsection covers Theorem 1 for bounded observables with pure point spectrum: EQ-MEAS, EQ-LIND and EQ-HS are equivalent, and under [B]FINENT EQ-KL is equivalent to EQ-MEAS.

Files.

- File: `briefs/ProjectiveMeasurement/ProjectiveMeasurement/InfiniteDimensional.lean`
- Library: `ProjectiveMeasurement`, which now builds every module `ProjectiveMeasurement.*`
- Imports:
  `ForPhyslib.QuantumMechanics.DephasingFlow`,
  `ForPhyslib.QuantumMechanics.VonNeumannRelativeEntropy`,
  and the finite-dimensional `ProjectiveMeasurement` (for `IsUniqueMinOn`)
- New file `ForPhyslib/QuantumMechanics/DiscreteObservable.lean`: resolutions of the identity and observables with pure point spectrum, the pinching as a strong sum, blocks and block-diagonal operators, positivity and trace of the pinching, and the Hilbert–Schmidt Pythagoras identity, through TauCeti's block splitting of the Hilbert–Schmidt energy (`TauCeti.Analysis.InnerProductSpace.HilbertSchmidt.Pythagoras`).
- New file `ForPhyslib/QuantumMechanics/DephasingFlow.lean`: the bounded generator and its exponential, the blocks of a solution, the unitary group e^{isX}, the Gaussian average and its trace-norm contraction, equilibration in trace norm.
- New file `ForPhyslib/QuantumMechanics/VonNeumannRelativeEntropy.lean`: eigenbases, the entropy as $\sum_i \eta(\lambda_i)$, the cross term $-\operatorname{tr}(\rho \log \sigma)$, Lindblad's formula, Klein's inequality, and the behaviour under pinching.

- New file `ForMathlib/Analysis/InnerProductSpace/CFCEigenvector.lean`: $g(T)v = g(\mu)v$ for $Tv = \mu v$, T self-adjoint and g continuous on the spectrum, in any dimension (TauCeti's version is finite-dimensional).

Setting. E is a complex Hilbert space of any dimension; separability is not used. Density operators, pure states and the entropy `vonNeumannEntropy` $\in [0, \infty]$ are as in the section "Hilbert spaces of any dimension" of `briefs/SchroedingerEvolution/formalization-log.tex`. An observable is $X \in \mathrm{End}(E)$ with `IsEigenDecomposition` X P x: the P_x are pairwise orthogonal projections with $\sum_x P_x = 1$ strongly, the x are distinct reals, and $XP_x = xP_x$; the index set is arbitrary. The pinching `IsResolutionOfIdentity.pinching` $\rho = \sum_x P_x \rho P_x$ is a strong sum, defined for every bounded ρ. The conditions are those of finite dimension, with these changes:

- `EqLind`: solutions are differentiable in operator norm for $t > 0$ and continuous from the right at 0; they tend to $\Phi(X, \rho)$ in trace norm.
- `EqHS`: minimization over `commutingDensityOperators` X, with TauCeti's `hilbertSchmidtNorm`.
- `EqKL`: only for ρ with $S(\rho)$ and $S(P_X(\rho))$ finite, and minimization over the density operators of finite entropy commuting with X. `EqMeasFinEnt` is EQ-MEAS on the same ρ.
- The relative entropy `relativeEntropy` is Lindblad's formula $\sum_{ij} |\langle e_i | f_j \rangle|^2 (\lambda_i \log \lambda_i - \lambda_i \log \mu_j + \mu_j - \lambda_i)$ in eigenbases of ρ and σ chosen once, each term in $[0, \infty]$ and $+\infty$ for $\mu_j = 0 < \lambda_i$.

Results.

Brief	Lean
Eq. (2), blocks determine ρ	`IsResolutionOfIdentity.ext_block`
Eqs. (3)–(4)	`IsEigenDecomposition.mul_dephasingGenerator_mul`
Eq. (5), for every solution	`IsEigenDecomposition.mul_eq_of_hasDerivAt`
A solution exists	`dephasingFlow`, `hasDerivAt_dephasingFlow`
$\rho(t)$ is a mixture of unitary conjugations of $\rho(0)$	`gaussianDephasing`, `IsEigenDecomposition.mul_gaussianDephasing_mul`, `IsEigenDecomposition.eq_gaussianDephasing`
Eqs. (6)–(10), trace-norm convergence	`IsEigenDecomposition.traceENorm_gaussianDephasing_sub_le`, `IsEigenDecomposition.tendsto_traceENorm_of_hasDerivAt`
EQ-LIND $\Leftrightarrow$ EQ-MEAS	`eqMeas_iff_eqLind`
"$[\sigma, X] = 0$ iff block diagonal iff $P_X(\sigma) = \sigma$"	`IsEigenDecomposition.commute_iff`, `IsResolutionOfIdentity.pinching_eq_self_iff`
P_X is positive and trace preserving	`IsResolutionOfIdentity.pinching_nonneg`, `posTrace_pinching`, `isDensityOperator_pinching`
Eq. (11)	`IsResolutionOfIdentity.hilbertSchmidtEnergy_sub`, `hilbertSchmidtNorm_sub_pinching_lt`
EQ-HS $\Leftrightarrow$ EQ-MEAS	`eqMeas_iff_eqHS`
$-\operatorname{tr}(\rho \log \sigma)$	`crossEntropy`, `IsEigenbasis.crossEntropy_eq`
$D(\rho\|\sigma) = -\operatorname{tr}(\rho \log \sigma) - S(\rho)$	`crossEntropy_eq_lindbladSum_add`, `relativeEntropy_add_vonNeumannEntropy`, `relativeEntropy_eq_lindbladSum`
"$\operatorname{tr}(\rho \log \sigma) = \operatorname{tr}(P_X(\rho) \log \sigma)$"	`IsResolutionOfIdentity.crossEntropy_pinching`
Eq. (12)	`IsResolutionOfIdentity.relativeEntropy_eq_add`
Eq. (13)	`IsResolutionOfIdentity.relativeEntropy_pinching`, `relativeEntropy_pinching_eq`
Klein, "vanishes iff $\sigma = P_X(\rho)$"	`eq_of_lindbladSum_eq_zero`, `relativeEntropy_eq_zero_iff`
EQ-KL $\Leftrightarrow$ EQ-MEAS under [B]FINENT	`eqKL_iff_eqMeasFinEnt`, via `IsResolutionOfIdentity.relativeEntropy_pinching_lt`
Theorem 1	`projective_measurement_tfae` and `eqKL_iff_eqMeasFinEnt`

Differences from the brief.

- Observables are bounded with pure point spectrum, given by their eigenprojections and distinct eigenvalues. The brief allows any self-adjoint X and remarks that the conditions fail on a continuous part of the spectrum; unbounded X, where the Lindblad equation holds only on a domain, is not formalized. Nor is a bridge from the spectral theorem: for a self-adjoint X with pure point spectrum, the P_x would be the spectral projections $\mu_S(\{x\})$.
- EQ-LIND is proved without the Hilbert–Schmidt estimate (6)–(9) and without the noncommutative Scheffé lemma. Every solution has the blocks of eq. (5), so it equals the Gaussian average $\int e^{isX} \rho\, e^{-isX}\, dN(0,t)(s)$, whose blocks carry the characteristic function $e^{-t(x-y)^2/2}$.

As an average of unitary conjugations, computed in $\mathcal{S}_1$, it is a trace-norm contraction. With $Q = \sum_{x \in F} P_x$ for a finite set F, $\|\rho(t) - P_X(\rho)\|_1$ is at most $\|\rho - Q\rho Q\|_1 + \|QP_X(\rho)Q - P_X(\rho)\|_1$, both small for large F uniformly in t, plus finitely many off-diagonal blocks of $Q\rho Q$, each decaying like $e^{-t(x-y)^2/2}$.

- The relative entropy follows decision D4: Lindblad's formula, in eigenbases chosen once. The cross term $-\operatorname{tr}(\rho \log \sigma)$ is defined without bases, as $\sup_n \operatorname{tr}(g_n(\sigma)\rho\, g_n(\sigma))$ for continuous g_n with $g_n^2 \uparrow -\log$. In any eigenbases, Lindblad's sum plus $S(\rho)$ equals the cross term, so the formula does not depend on the bases once $S(\rho) < \infty$; for $S(\rho) = \infty$ independence is not formalized. Klein's equality case is read off the sum: a vanishing term forces $\lambda_i = \mu_j$ or $\langle e_i | f_j \rangle = 0$, so ρ and σ agree on every $\langle e_i | \cdot f_j \rangle$. Agreement with the finite-dimensional `relEntropy` is not formalized.

- The Pythagoras identity (12) is proved in the form $D(\rho\|\sigma) + S(\rho) = -\operatorname{tr}(P_X(\rho) \log \sigma) = D(P_X(\rho)\|\sigma) + S(P_X(\rho))$. The middle step holds because each $g_n(\sigma)$ commutes with every P_x and the pinching preserves traces. No eigenbasis adapted to the blocks is needed.

- Under [B]FINENT the hypothesis $S(\rho) < \infty$ is redundant: $S(\rho) \le S(P_X(\rho))$ follows from the identities above. `EqKL` keeps it, as the brief does.

- The Hilbert–Schmidt Pythagoras identity holds in $[0, \infty]$ for every operator, through TauCeti's block splitting of the energy, with no summability assumption. The strict inequality uses that differences of density operators have finite energy.

Problems found in the brief. No errors in the statements. Gaps beyond those of the finite-dimensional part:

- "Both $\rho(t)$ and $P_X(\rho(0))$ are density operators" is asserted. In infinite dimension it is needed for the Scheffé step. The Gaussian representation above proves it, and also gives the contraction that replaces that step.

- The Scheffé step itself ("Hilbert–Schmidt convergence to a density operator implies trace-norm convergence") is cited without reference or proof; it is not needed here.

- The brief solves the block equations (4) but does not say that the solution is unique; the argument behind `IsEigenDecomposition.mul_eq_of_hasDerivAt` gives uniqueness for every solution continuous from the right at 0.

- Eq. (13)'s "finite if and only if $S(P_X(\rho))$ is finite" uses $S(\rho) < \infty$, which the admissible domain provides; outside it the equivalence fails, since a block-diagonal ρ with $S(\rho) = \infty$ has $D(\rho\|P_X(\rho)) = D(\rho\|\rho) = 0$. The remark "Relative entropy in infinite dimensions" says that in the case $S(P_X(\rho)) = +\infty$ "every ensemble commuting with X is infinitely distant from ρ". Read literally, the same ρ is a counterexample, with $\sigma = \rho$. The claim holds when $S(\rho) < \infty$, by $D(\rho\|\sigma) = -\operatorname{tr}(P_X(\rho) \log \sigma) - S(\rho) \ge S(P_X(\rho)) - S(\rho)$; this is not formalized.